Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

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The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Daniel Lewis (Principal Investigator) Chaitanya Ullal (Co-Principal Investigator) Semih Akin (Co-Principal Investigator)	**Email**	lewisd2@rpi.edu
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2. Technology Overview

Title of Proposed Technology	Economically Accessible Functionally Graded Products through Laser Powder Fusion Additive Methods			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Powder	Powder Processing Technologies	Metal Materials	Materials Processing Technologies
Concept of Proposed Technology	This technological innovation uses our ability to deliver chemical composition variation within additive powder processing to create metallic components with functionally optimized material distributions, placing oxidation- and corrosion-resistant alloys at the surface while maintaining structurally superior, high-strength materials in the interior. Use of the Formalloy and Aloy Development Feeder we can spatially tailor composition, to directly address the conflicting requirements of environmental exposure and mechanical performance without relying on discrete coatings or bonded interfaces that are prone to delamination, cracking, or diffusion-related degradation. The graded or multi-material architecture enhances component lifetime in aggressive thermal and chemical environments, improves damage tolerance under cyclic loading, and enables weight and cost reduction by reserving expensive, high-alloy or noble materials only where they are needed. Compared to conventional manufacturing routes for products that meet the customizability and complexity threshold for AM processing, this strategy expands the design space for metallic systems,			

	supports design for manufacturing through part consolidation, and offers a scalable pathway to high-performance components for aerospace, energy, chemical processing, and industrial applications. **Using industrially suggested component designs and bulk alloys**, our team will apply fundamental knowledge of microstructure science, alloy phase diagrams, physical metallurgy, and diffusion kinetics to design compositional pathways that smoothly transition between dissimilar interior and exterior alloys while maintaining performance. By selecting intermediate compositions that avoid brittle phases, promote microstructural compatibility between regions, avoid excessive segregation, or steep thermal expansion mismatches, the approach ensures microstructural stability and mechanical integrity across the graded region. Our understanding of diffusion behavior during processing and in-service operation permits our design and understanding of interfacial degradation mechanisms, residual stress, and long-term property drift. Together, these metallurgy-informed pathways enable robust, manufacturable multi-material metallic components with predictable performance.
Rationale for Joint R&D Project	• Academic expertise in phase diagrams, diffusion, and microstructural evolution complements industry's understanding of manufacturability, cost, certification, and performance requirements, ensuring solutions are both scientifically sound and industrially viable. • Industry partners provide production-scale additive powder systems, relevant non-proprietary model alloys, and realistic service environments, while academic partners contribute advanced characterization and manufacturing tools (electron microscopy, thermal analysis, dilatometry, Formalloy Laser Fusion system, etc.) not typically available at industrial partner locations. • Functionally graded, multi-material metallic systems involve technical and qualification risks that are more effectively shared between academia and industry. • Industry-defined use cases will guide our lab's implementation efforts toward relevant performance targets, reducing the time required to transition discoveries into industry relevant components and alloys. • Long-term testing under realistic thermal, mechanical, and corrosive service conditions are more easily accessible to industry partners.
Key Words	1. Functionally Graded Metals 2. Multi-material Additive Manufacturing 3. Oxidation and Corrosion Resistance 4. Diffusion Engineered Materials 5. Microstructure-Property Design

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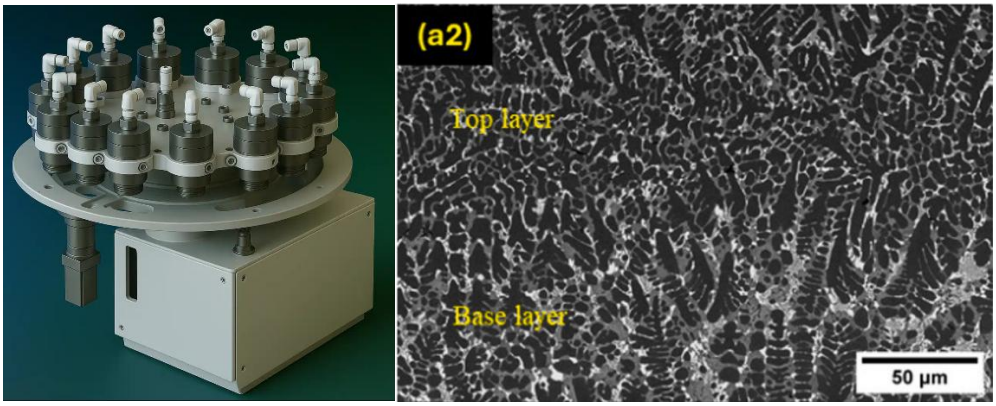
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level ( )	End of the Project	Level ( )
R&D Objectives	• **Research Objective**: Identify and optimize additive manufacturing process parameters and compositional gradients that produce stable, defect-tolerant microstructures across functionally graded metallic components, with particular emphasis on controlling phase formation, residual stress, and diffusion behavior at material transitions. • **Development Objective**: Establish validated process–structure–property relationships linking processing conditions and alloy chemistries to microstructural stability and degradation mechanisms, enabling quantitative in-service life prediction models for oxidation, corrosion, and mechanically loaded environments.			
R&D Target Specifications	**Target Performance Specifications** **Interfacial integrity and defect density**: This will be quantified by measurement of porosity, the number of observable fusion defects, and crack density across graded regions, with targets defined relative to monolithic industry baseline materials such as stainless steels and nickel-based alloys. **Microstructural stability under service conditions**: Measured by phase stability through observation of characteristic microstructural length scales, grain coarsening rates through determination of interfacial or diffusion-based control mechanisms, and diffusion zone growth and inter-alloy mixing after thermal exposure using thermal cycling representative of in-service environments.  **Figure 1 Left.** The RPI Formalloy system includes an Alloy Development Feeder (ADF) that will permit design, development and fabrication of metallic FGMs for this project. **Figure 1 Right.** This program leverages our experience in developing high entropy alloys (HEAs) as intermediate layers in bond coat applications. Our team understands the phase formation sequences upon solidification and the phase chemistry throughout the gradient region. The alloy microstructure shown is from an HEA bondcoat application recently published in the Journal of Manufacturing Processes. **Environmental resistance of the graded surface**: Evaluated through oxidation kinetics of surface materials as compared to the bulk, corrosion rates, and scale adherence of surface regions compared to conventional coatings or bulk corrosion-resistant alloys. **Mechanical performance across the gradient**: Assessed via phase-specific microhardness, interfacial and phase strength through examination of hardness indent damage ensuring smooth property transitions without stress-concentrating discontinuities.			

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	Predictability for life assessment models: The team has already demonstrated by the ability to parameterize models of diffusion, oxidation, and corrosion, and also estimated damage evolution behavior using experimentally derived process–structure–property data from the graded material in previous work. We will apply those skills to this new set of applications.
R&D Plan	This R&D plan establishes a metallurgy-informed framework for designing and manufacturing oxidation- and corrosion-resistant surfaces with structurally optimized interiors using additive powder processing. The resulting process maps and stability data will enable scalable, reliable deployment of functionally graded metallic components in demanding industrial environments. **Phase 1: Powder Sourcing, and Compositional Design** **Objective**: Enable controlled compositional gradients using industrially relevant or specified powders compatible with multi-material additive processing. **Activities**: • Source commercial oxidation- and corrosion-resistant surface alloy powders and high-strength structural interior alloy powders. • Synthesize or modify powders (e.g., blending, pre-alloying, particle size tailoring) to create intermediate compositions informed by alloy phase diagrams. • Characterize powder chemistry, morphology, flowability, and oxygen content. **Deliverables** The team will deliver: 1) experimental powder chemistries for interior, exterior, and graded transition regions; 2) powders will be characterized using electron microscopy and stereological techniques; and 3) initial compositional pathway maps linking different alloy systems. These maps will be delivered using binary or pseudo-binary or pseudo-ternary phase diagrams. Where possible, linkages between key elements will be presented on binary phase diagrams to facilitate the design process. **Phase 2: Fabrication of Multilayer and Graded Structures** **Objective**: Demonstrate additive manufacturing of discrete multilayers and continuous composition gradients. **Activities**: • Fabricate multilayer specimens with controlled layer thickness and composition using additive powder processing. • Transition from stepwise multilayers to continuously graded compositions. • Optimize process parameters (energy density, scan strategy, powder feed ratios) for metallurgical bonding and defect minimization. **Deliverables** For components fabricated on the Alloy Development Feeder we will deliver defect-minimized multilayer and graded metallic specimens for oxidation, corrosion and microstructure stability studies. We will document preliminary processing–composition relationships and begin mapping phase diagrams and pseudo-phase diagrams that will be used for design of intermediate compositions.

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Phase 3: Microstructural and Chemical Characterization

Objective: Quantify phase evolution, chemical gradients, and interfacial integrity of specimens.

Activities

- Characterize microstructures using SEM/EDS, EBSD, XRD, and TEM as appropriate.
- Measure chemical gradients, diffusion profiles, and phase distributions across layers.
- Identify formation of undesirable phases, segregation, or residual porosity.
- Downselect FGM designs based on industry input or simulated in-service performance.

Deliverables

High-resolution microstructure and chemistry maps will be developed across graded regions to quantify phase distributions, grain structure, defect populations, and elemental gradients as a function of position and processing conditions. These datasets will enable direct correlation between local composition, thermal history, and resulting microstructural features within functionally graded materials. Using these maps, stable compositional transitions that maintain desirable phases and mechanical continuity will be distinguished from unstable transitions that promote brittle phase formation, excessive diffusion, or microstructural degradation. The resulting understanding will inform the design of robust grading strategies that support long-term performance and predictable in-service behavior.

Phase 4: Microstructural Stability and Performance Assessment

Objective: Evaluate stability under thermal, mechanical, and environmental exposure.

Activities

- Conduct thermal exposure and thermal cycling experiments.
- Assess diffusion-driven evolution, phase transformations, and residual stress relaxation through phase specific hardness indentations.
- Evaluate oxidation and corrosion behavior of graded surfaces relative to monolithic alloys.

Deliverables

Stability maps linking composition, temperature, and time. Through characterization like the previous phase, we will develop data on degradation mechanisms and service-relevant performance.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The academic research institute will be responsible for implementing and optimizing multi-material additive manufacturing using its laser powder bed fusion system equipped with an alloy development feeder. This capability enables precise, layer-by-layer and within-layer control of alloy composition, allowing systematic exploration of compositionally graded alloys and diffusion-engineered transitions informed by physical metallurgy. Rensselaer will be responsible for conducting the fundamental materials science investigations required to design, characterize, and validate functionally graded metallic systems for coupon and prototype fabrication. This includes advanced electron beam–based characterization capabilities such as scanning electron microscopy (SEM), electron backscatter diffraction (EBSD), transmission electron microscopy (TEM), and energy-dispersive X-ray spectroscopy (EDS) to resolve microstructural features, phase distributions, crystallographic texture (where necessary), and chemical gradients at high spatial resolution. These techniques will be used to quantify interfacial integrity, diffusion behavior, stereologically measured defect populations, and phase stability (through thermal exposure) across graded regions. The PIs' expertise in electron beam characterization ensures the microstructural understanding needed to support design and life prediction.
Expected R&R of Korean SME Partner	We ask that the industrial partner support Rensselaer research by defining application-driven use cases and performance requirements, provide access to industrially relevant additive manufacturing equipment to test our ideas on site, recommend alloy powders, and non-proprietary processing knowledge, and sharing materials data and practical insights on degradation, qualification, and scale-up challenges. Our partner would enable realistic validation through representative in-service testing of oxidation, corrosion, thermal cycling, and mechanical performance, while guiding alignment with certification and standards requirements particular to their industrial applications. We expect periodic participation in technical reviews and if possible, student mentorship to ensure that fundamental research outcomes translate efficiently into manufacturable, deployable functionally graded metallic components.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes of this collaborative project include advancing FGM technology and increasing commercialization readiness. Technologically, the project will deliver validated methods for producing functionally graded metallic components using additive powder processing, including metallurgy-informed compositional pathways, stable graded microstructures, and predictive process–structure–property relationships that support in-service life calculations. These outcomes will ultimately reduce technical risk for multi-material additive manufacturing at our industrial partner.

6. References

(1) Compositional variations and oxidation characteristics of Al-Cr-Fe-Ni based high entropy alloys deposited via twisted-wire-arc surfacing; Joni C. Dhar, Heather Murdoch, Pauline Smith, Sandipan Mishra, Daniel Lewis, Johnson Samuel, Journal of Manufacturing Processes 141 (2025) 1675–1689

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization